

HIVA OA, French Polynesia

WMO: 919250

Lat: 9.806S

Lon: 139.036W

Elev: 53

StdP: 100.69

Time Zone: -9.50 (MRQ)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.6	22.0	16.3	11.7	25.7	17.6	12.7	25.6	9.4	26.8	8.7	27.1	0.9	30	0.321

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	6.4	32.4	25.3	31.7	25.0	31.2	24.8	26.6	30.5	26.2	30.1	25.8	29.7	3.7	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.6	20.9	29.1	25.1	20.3	28.6	24.7	19.8	28.3	83.7	30.7	81.7	30.2	80.2	29.8	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.7	8.0	7.2	DB	20.8	33.7	0.9	0.9	20.1	34.3	19.6	34.9	19.1	35.4	18.4	36.0	(4)
(5)			WB	18.4	27.2	0.9	0.7	17.7	27.7	17.2	28.1	16.6	28.4	15.9	28.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	26.6	26.9	27.2	27.4	27.3	26.8	26.1	25.7	25.5	25.9	26.2	26.6	26.9	(6)
(7)		DBStd	1.03	0.88	0.92	0.90	0.84	0.76	0.77	0.75	0.68	0.81	0.83	0.87	0.90	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	6041	524	482	540	518	521	484	486	482	477	503	499	525	(10)
(11)		CDD18.3	2999	266	249	282	268	262	234	228	223	227	244	249	267	(11)
(12)		CDH23.3	25999	2481	2399	2765	2594	2301	1856	1652	1564	1752	1977	2166	2492	(12)
(13)		CDH26.7	7027	783	790	874	768	603	358	244	251	357	528	665	804	(13)
(14)	Wind	WSAvg	2.8	2.9	2.8	2.6	2.4	2.3	2.6	2.9	2.9	3.1	2.9	2.7	2.9	(14)
(15)	Precipitation	PrecAvg	1298	107	97	155	155	131	166	135	118	73	84	74	81	(15)
(16)		PrecMax	2312	544	265	604	466	401	587	357	343	214	300	202	252	(16)
(17)		PrecMin	560	17	7	11	12	17	40	40	13	14	7	3	7	(17)
(18)		PrecStd	421	98	61	123	125	90	102	75	73	46	65	48	59	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.6	33.0	32.9	32.7	32.0	30.9	30.2	30.3	31.1	32.1	32.5	32.6	(19)
(20)			MCWB	24.9	25.4	25.9	25.9	25.2	24.6	24.0	23.9	23.9	24.8	25.1	24.8	(20)
(21)		2%	DB	31.6	32.0	31.9	31.7	31.0	30.0	29.2	29.3	30.0	31.0	31.3	31.6	(21)
(22)			MCWB	24.4	25.2	25.7	25.7	25.2	24.5	23.9	23.7	23.7	24.1	24.4	24.5	(22)
(23)		5%	DB	30.8	31.1	31.1	30.8	30.2	29.2	28.6	28.7	29.2	30.1	30.6	30.9	(23)
(24)			MCWB	24.3	25.0	25.5	25.4	25.1	24.3	23.8	23.5	23.6	23.9	24.1	24.2	(24)
(25)		10%	DB	30.1	30.4	30.3	30.1	29.5	28.6	28.0	28.1	28.6	29.3	29.8	30.2	(25)
(26)			MCWB	24.1	24.8	25.3	25.2	24.9	24.1	23.6	23.4	23.5	23.7	23.9	24.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.7	27.1	27.2	27.2	26.7	25.8	25.2	25.1	25.1	25.8	26.2	26.2	(27)
(28)			MCDWB	30.4	31.0	31.1	30.8	30.0	29.0	28.4	28.6	29.3	30.3	30.7	31.1	(28)
(29)		2%	WB	25.7	26.2	26.5	26.4	26.0	25.2	24.7	24.5	24.6	25.1	25.4	25.5	(29)
(30)			MCDWB	29.7	30.3	30.4	30.1	29.5	28.4	27.9	28.0	28.6	29.5	29.9	30.2	(30)
(31)		5%	WB	25.1	25.7	26.0	26.0	25.6	24.9	24.3	24.1	24.2	24.6	24.9	25.0	(31)
(32)			MCDWB	29.3	29.8	30.0	29.7	29.1	28.0	27.4	27.6	28.1	28.8	29.3	29.6	(32)
(33)		10%	WB	24.7	25.2	25.6	25.6	25.2	24.5	24.0	23.7	23.9	24.2	24.5	24.5	(33)
(34)			MCDWB	28.8	29.3	29.4	29.2	28.7	27.6	27.1	27.1	27.7	28.3	28.7	29.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.7	6.7	6.4	6.0	5.9	5.3	5.1	5.5	5.9	6.7	6.9	6.9	(35)
(36)			MCDBR	7.7	7.8	7.3	7.2	7.0	6.3	6.0	6.5	6.8	7.9	7.9	7.8	(36)
(37)			MCWBR	3.2	3.5	3.4	3.3	3.4	3.1	3.0	3.2	3.2	3.5	3.4	3.3	(37)
(38)		5% WB	MCDWB	6.8	6.8	6.6	6.3	6.3	5.6	5.4	5.8	6.0	6.9	7.1	6.9	(38)
(39)			MCWBR	3.4	3.4	3.4	3.3	3.4	3.1	2.9	3.1	3.1	3.5	3.5	3.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.373	0.372	0.373	0.380	0.369	0.366	0.364	0.365	0.372	0.363	0.366	0.367	(40)	
(41)		taud	2.594	2.614	2.621	2.611	2.635	2.623	2.621	2.605	2.576	2.606	2.607	2.611	(41)	
(42)		Ebn at Noon	968	963	946	910	892	879	888	912	932	960	967	972	(42)	
(43)		Edh at Noon	104	102	100	96	89	88	89	95	103	102	102	102	(43)	
(44)	All-Sky Solar Radiation	RadAvg	7.08	7.12	6.66	6.16	5.60	5.26	5.42	6.01	6.62	7.01	7.03	7.11	(44)	
(45)		RadStd	0.44	0.27	0.38	0.38	0.20	0.12	0.18	0.16	0.16	0.25	0.31	0.28	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	+0.57	N/A	N/A	N/A	N/A	+141	+141
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page